

Gabriel Cha

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EDUCATION

Columbia University

M.S. in Data Science, 3.53 GPA

New York, NY

09/2025 – 12/2026

- Applied Large Language Models & Agentic AI, Distributed Compute & LLM Training, Deep Learning & Neural Networks

University of California, San Diego

B.S. in Data Science, 3.75 GPA

San Diego, CA

09/2021 – 05/2025

- Teaching Assistant: Taught data science (DSC 10, DSC 80) for 9 quarters to 1,200+ students, holding 120+ office hours

EXPERIENCE

Honeywell

AI/ML Engineer Intern

Atlanta, GA

05/2026 – 08/2026

- Building `llms.txt`, a knowledge harness for Honeywell's security-platform API, so AI coding agents (Cursor, Claude Code) generate accurate REST calls without hallucinating parameters or auth flows, reducing token usage by 60%
- Built a context graph of the platform's physical entities (sites, cameras, fobs, people, access schedules), giving AI agents structural understanding of how they relate to chain API calls correctly
- Prototyping agentic, self-aware security workflows on top of the harness that read live API feeds and triage which of hundreds of camera feeds an operator should prioritize

Root Access AI (\$2M venture-backed startup)

AI/ML Engineer Intern

New York, NY

09/2025 – 05/2026

- Engineered autonomous AI coding agents in LangChain tailored for firmware, transforming hardware datasheets into accurate device-control code; on a 5-person team, grew the enterprise customer base from 3 to 7
- Ensured precise alignment with user intent by developing a "Plan Mode" that empowers the agent to autonomously run grep commands and explore the architecture before writing any code
- Reduced user-correction turns by 40% by building an automated feedback loop where the agent runs C build commands to test its own work and self-correct compilation errors on the fly
- Increased the agent's hardware-specific accuracy by 34% by fine-tuning open-weight LLMs via QLoRA on cloud A100s using a custom-curated dataset

DSTL Lab

AI Research Assistant

San Diego, CA

10/2024 – 05/2025

- Engineered an AI agent that generates course-aligned practice problems by retrieving and processing course-specific materials against active learning objectives; co-authored a SIGCSE 2026 paper on the work
- Raised LLM generation accuracy from 23% to 87% by designing a two-stage data pipeline that cleaned and structured raw educational content into optimized model inputs

Edison

Data Science Intern

Los Angeles, CA

06/2024 – 09/2024

- Built a scalable system monitoring 70+ solar plants for automated outlier detection, transforming a 3-day manual review into a 10-minute automated workflow and preventing \$5K+/month in losses from undetected faults
- Developed a Python ETL pipeline processing 1TB+ across 100+ files, eliminating 4-day manual reporting cycles to accelerate data delivery for energy-trading clients

PROJECTS

Cloi – AI Coding Agent (400 GitHub Stars)

05/2025

- Built an open-source agent that diagnoses runtime errors, retrieves relevant code via RAG, and applies LLM-generated fixes directly in the codebase; invited to YC S25 interview

Classifying Emotions with Machine Learning

09/2023 – 06/2024

- Led \$6,500 grant-funded research classifying emotions in human speech, training and benchmarking multiple machine learning models to identify the top performer

SKILLS

AI & Agents: LangChain, LangGraph, LLMs, RAG, Fine-Tuning (QLoRA), Context Engineering, LLM-as-a-Judge

Machine Learning: PyTorch, TensorFlow, Hugging Face, Scikit-learn, NLP, Distributed Training

Languages & Data: Python, SQL, Java, Git, Pandas, Spark, NumPy, Tableau, ETL, A/B Testing